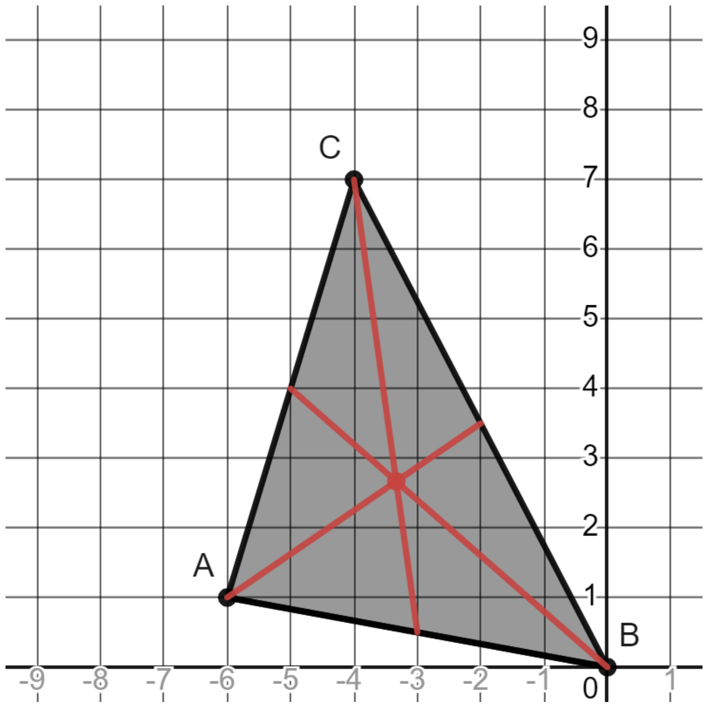
Geometry
Unit 13
Lesson 7 Homework

Name___Answer Key_______
Date____________________
Period___________________

Use the following information for questions 1-4. Triangle ABC has vertices at $A(-6,1)$, $B(0,0)$, and $C(-4,7)$.



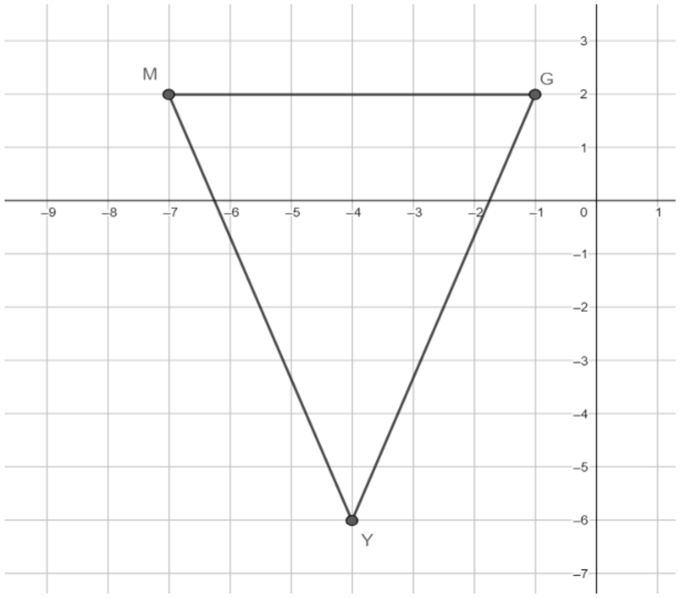
1. What is the average of the x-coordinates? Round to the nearest thousandth if necessary.
$\frac{-6+0-4}{3} = -\frac{10}{3} \approx -3.333$

2. What is the average of the y-coordinates? Round to the nearest thousandth if necessary.
$\frac{1+0+7}{3} = \frac{8}{3} \approx 2.667$

3. What are the coordinates of the centroid?
$(-3.333, 2.667)$

4. Verify the coordinates of the centroid by drawing the median of each side.
See graph.

Use the following information for question 5. Triangle GMY has vertices at $G(-1,2)$, $M(-7,2)$, and $Y(-4,-6)$.



5. What are the coordinates of the centroid of triangle GMY?

$\left(\frac{-1-7-4}{3}, \frac{2+2-6}{3}\right) = \left(-4, -\frac{2}{3}\right)$

6. $\triangle KLM$ has vertices at $K(7,2)$, $L(6,9)$, and a centroid at $(5,6)$. What are the coordinates of vertex M?

$\frac{7+6+M_x}{3} = 5$

$13 + M_x = 15$

$M_x = 2$

$\frac{2+9+M_y}{3} = 6$

$11 + M_y = 18$

$M_y = 7$

$(2,7)$

Keziah is finding the centroid of $\triangle ALG$ with vertices at $A(-1,3)$, $L(2,0)$, and $G(-4,-6)$.

Use the following information for questions 7-8. Her work is shown.

Keziah's Work	
$x = \frac{1+2+4}{3}$ $x = \frac{7}{3}$	$y = \frac{3+0+6}{3}$ $y = \frac{9}{3} = 3$
Centroid $\left(\frac{7}{3}, 3\right)$	

7. What is Keziah's mistake?

She changed all negative values to positive (A_x, G_x, G_y) before calculating the averages.

8. What are the coordinates of the centroid of $\triangle ALG$?

$\left(\frac{-1+2-4}{3}, \frac{3+0-6}{3}\right) = (-1, -1)$